

Properties of Parallelograms

Jonathan R. Bacolod

A **parallelogram** is a quadrilateral with both pairs of opposite sides parallel.

Opposite Sides

Opposite sides of a parallelogram are congruent.

$$AB = CD \quad \text{and} \quad BC = AD$$

Opposite Angles

Opposite angles of a parallelogram are congruent.

$$\angle A = \angle C \quad \text{and} \quad \angle B = \angle D$$

Consecutive Angles

Consecutive angles are supplementary; their measures add up to 180° .

$$\angle A + \angle B = 180^\circ$$

Diagonals

The **diagonals** of a parallelogram bisect each other.

If diagonals AC and BD intersect at M , then

$$AM = MC \quad \text{and} \quad BM = MD$$

Finding Angles

If one angle of a parallelogram is known, the other angles can be determined using its properties.

For example, if

$$\angle A = 70^\circ,$$

then

$$\angle C = 70^\circ$$

and

$$\angle B = \angle D = 110^\circ.$$

Congruent Diagonal Segments

If the diagonals intersect at a point, each diagonal is divided into two congruent segments.

$$AM = MC \quad \text{and} \quad BM = MD$$

Examples

Example 1

In parallelogram $ABCD$, $AB \parallel CD$ and $BC \parallel AD$.

What property is illustrated?

Both pairs of opposite sides are parallel.

Example 2

In parallelogram $ABCD$, $AB = CD$ and $BC = AD$.

What property is illustrated?

Opposite sides are congruent.

Example 3

In parallelogram $ABCD$, $\angle A = \angle C$ and $\angle B = \angle D$.

What property is illustrated?

Opposite angles are congruent.

Example 4

In parallelogram $ABCD$, $\angle A + \angle B = 180^\circ$.

What property is illustrated?

Consecutive angles are supplementary.

Example 5

In parallelogram $ABCD$, diagonals AC and BD intersect at M and $AM = MC$.

What property is illustrated?

The diagonals bisect each other.

Practice Exercises

Practice 1

In parallelogram $ABCD$, $AB \parallel CD$ and $BC \parallel AD$.

Identify the property of parallelograms illustrated by the statement.

Practice 2

In parallelogram $ABCD$, $AB = CD$ and $BC = AD$.

Identify the property of parallelograms illustrated by the statement.

Practice 3

In parallelogram $ABCD$, $\angle A = \angle C$ and $\angle B = \angle D$.

Identify the property of parallelograms illustrated by the statement.

Practice 4

In parallelogram $ABCD$, $\angle A + \angle B = 180^\circ$.
Identify the property of parallelograms illustrated by the statement.

Practice 5

In parallelogram $ABCD$, $\angle C + \angle D = 180^\circ$.
Identify the property of parallelograms illustrated by the statement.

Practice 6

In parallelogram $ABCD$, diagonals AC and BD intersect at M and $AM = MC$.

Identify the property of parallelograms illustrated by the statement.

Practice 7

In parallelogram $ABCD$, diagonals AC and BD intersect at M and $BM = MD$.

Identify the property of parallelograms illustrated by the statement.

Practice 8

In parallelogram $ABCD$, $\angle A$ and $\angle D$ form a supplementary pair.

Identify the property of parallelograms illustrated by the statement.

Practice 9

In parallelogram $ABCD$, opposite sides have the same length.

Identify the property of parallelograms illustrated by the statement.

Practice 10

In parallelogram $ABCD$, the diagonals divide each other into two congruent segments.

Identify the property of parallelograms illustrated by the statement.

Activities

Activity 1

In parallelogram $WXYZ$, $WX \parallel YZ$ and $XY \parallel WZ$.

Identify the property of parallelograms illustrated by the statement.

Activity 2

In parallelogram $WXYZ$, $WX = YZ$ and $XY = WZ$.

Identify the property of parallelograms illustrated by the statement.

Activity 3

In parallelogram $WXYZ$, $\angle W = \angle Y$ and $\angle X = \angle Z$.

Identify the property of parallelograms illustrated by the statement.

Activity 4

In parallelogram $WXYZ$, $\angle W + \angle X = 180^\circ$.
Identify the property of parallelograms illustrated by the statement.

Activity 5

In parallelogram $WXYZ$, $\angle Y + \angle Z = 180^\circ$.
Identify the property of parallelograms illustrated by the statement.

Activity 6

In parallelogram $WXYZ$, diagonals WY and XZ intersect at N and $WN = NY$.

Identify the property of parallelograms illustrated by the statement.

Activity 7

In parallelogram $WXYZ$, diagonals WY and XZ intersect at N and $XN = NZ$.

Identify the property of parallelograms illustrated by the statement.

Activity 8

In parallelogram $WXYZ$, $\angle X$ and $\angle Y$ are supplementary.

Identify the property of parallelograms illustrated by the statement.

Activity 9

In parallelogram $WXYZ$, opposite sides are congruent.

Identify the property of parallelograms illustrated by the statement.

Activity 10

In parallelogram $WXYZ$, the diagonals bisect each other.

Identify the property of parallelograms illustrated by the statement.

Thank you!